

# Is Data-Reuploading Really a Cheat Code? An Experimental Analysis

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**Abstract:** Data Reuploading has been proposed as a generic embedding strategy in Variational Quantum Circuits (VQCs), offering a systematic approach to encoding classical data without the need for problem-specific circuit design. Prior studies have suggested that increasing the number of reuploading layers enhances model performance, particularly in terms of expressibility. In this paper, we present an experimental analysis of Data Reuploading, systematically evaluating its impact on expressibility, trainability, and completeness in classification tasks. Our results indicate that while adding some reuploading layers can improve performance, excessive layering does not lead to expressibility gains and introduces barren plateaus, significantly hindering trainability. Consequently, although Data Reuploading can be beneficial in certain scenarios, it is not a “cheat code” for optimal quantum embeddings. Instead, the selection of an effective embedding remains an open problem, requiring a careful balance between expressibility and trainability to achieve robust quantum learning models.

## 1 INTRODUCTION

Quantum computing has witnessed tremendous growth in recent years and is rapidly being recognized as the next frontier for optimization and, possibly, machine learning (ML) applications. By leveraging distinct quantum phenomena, such as superposition and entanglement, potentially transformative speed-ups and improvements over classical approaches are anticipated (Zhou et al., 2020). Nonetheless, the field remains in its early stages, and realizing its full potential often requires a high level of specialized knowledge, extensive parameter tuning, and careful error mitigation (Khanal et al., 2024). Consequently, building and training quantum models that are both robust and effective still poses significant challenges.

In response, recent studies have emerged reporting promising strategies for simplifying quantum-based ML model construction (Schuld et al., 2021). One such strategy is *Data Reuploading*, which has shown potential for improving how classical information is embedded into quantum circuits (Pérez-Salinas et al., 2020). In this paper, we aim to provide an experimental analysis of Data Reuploading, exploring both its

capabilities and its limitations. Through systematic experimentation, we seek to study the practical benefits of this technique, while highlighting key challenges that remain in this field.

### 1.1 Quantum Variational Machine Learning

A core approach to Quantum Machine Learning (QML) involves the use of *Variational Quantum Circuits* (VQCs) (Benedetti et al., 2019). In this paradigm, a parametrized quantum circuit is trained via classical optimization techniques (e.g., gradient descent) to solve a variety of learning tasks. Although QML methods can be broadly divided into kernel-based and VQC-based approaches (Mengoni and Di Pierro, 2019), the latter has gained special prominence due to its potential flexibility and expressive power.

VQC-based models typically consist of tunable gates that act on qubits, altering their initial quantum state. This initial state is usually obtained by encoding classical data using additional gates. Once these gates are parametrised, a cost function is defined to measure performance on the task at hand (e.g., classification error or policy gradient in reinforcement learning). A classical optimizer iteratively updates the circuit’s pa-

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rameters to minimize this cost. Despite the promise of VQC-based methods, key challenges persist. Issues such as *barren plateaus*, where gradients vanish exponentially as circuit depth increases (McClean et al., 2018; Sim et al., 2019), and the intrinsic noise of near-term quantum devices continue to limit the scalability and reliability of these approaches.

## 1.2 Data Reuploading

Among the techniques proposed to overcome some of these limitations, *Data Reuploading* has garnered significant attention. Originally introduced by Pérez-Salinas *et al.* (Pérez-Salinas et al., 2020), Data Reuploading involves encoding the same classical input data multiple times throughout the VQC, rather than just once at the beginning. This approach bears resemblance to other VQC-related works such as the Quantum Approximate Optimization Algorithm (QAOA), which promises eventual convergence onto the optimal result (Blekos et al., 2024).

A key theoretical justification for why such repeated angle-rotation blocks remain expressive comes from the work of Schuld *et al.* By injecting data angles multiple times throughout the circuit, one effectively generates higher-order Fourier terms in the model’s expansion, thereby enabling complex function approximation (Schuld et al., 2021). Hence, the circuit can realize a rich variety of decision boundaries, even if each individual layer relies on a relatively small number of parameters.

The method was originally created as part of a Universal Quantum Classifier for generic ML tasks. However, subsequent works have highlighted its potential to substantially simplify the classical data embedding process into the quantum circuit, avoiding the need to generate complex or tailored strategies, as it is the case with other well-known techniques such as the Variational Quantum Eigensolver (VQE) (Peruzzo et al., 2014; Tilly et al., 2022).

Early empirical studies have shown promising results, indicating that repeated encoding can improve performance in tasks such as classification and reinforcement learning (Lan, 2021; Skolik et al., 2022; Kölle et al., 2024), making the technique an attractive avenue of research for building more generalizable quantum ML models.

Still, much remains to be understood about the trade-offs associated with layering additional encoding blocks. While these layers may bolster expressibility, they may also amplify trainability issues like barren plateaus and noise sensitivity (Coelho et al., 2024). Through the experimental analysis presented in this paper, we aim to clarify how Data Reupload-

ing impacts the trainability, expressibility, and completeness of VQCs in realistic ML tasks. Our goal is to identify scenarios where the repeated encoding strategy provides clear benefits, as well as the limitations that must be addressed to achieve more robust and scalable quantum ML models.

## 2 METHODOLOGY

In this section, we outline our methodological framework for examining Data Reuploading in Variational Quantum Circuits. Our study focuses on a binary classification task using a synthetic dataset, where we test multiple configurations of Data Reuploading. We evaluate the resulting circuits along three key dimensions: *trainability*, *expressibility*, and *completeness*, which will be discussed in more detail in Section 2.3. By combining systematic experimentation with these evaluative criteria, we aim to gain a clear understanding of both the benefits and limitations of Data Reuploading in a practical ML setting (Holmes et al., 2022).

### 2.1 Generation of the Dataset

To investigate the performance of different Data Reuploading configurations, we generate a synthetic dataset using the `make_classification()` function from `scikit-learn` (Pedregosa et al., 2011). Specifically, we create a binary classification dataset consisting of 200 samples, with 5 informative features and no redundant attributes, organised in 2 clusters per class. Each feature is then scaled by a factor of 10 before being passed through the transformation  $2 \cdot \arctan(x)$ . This scaling amplifies the separation between points in the feature space and mitigates issues associated with mapping values near zero via the arctan function, ultimately helping to improve the distinctiveness of the embedded data in our VQC. A random seed is set to ensure reproducibility in the results.

We chose 200 rows, following the general rule of thumb that a classification task requires at least as many rows as ten times the number of features for each class, and doubled that amount for additional confidence.

### 2.2 Experimental Setup

#### 2.2.1 Data Reuploading Circuit

In the original Data Reuploading framework, classical data  $\mathbf{x}$  are repeatedly introduced into a quantum

circuit through a data-encoding gate  $U(\mathbf{x})$ , followed by a trainable gate  $U(\theta)$  (Pérez-Salinas et al., 2020).

$$L(i) \equiv U(\theta_i)U(\mathbf{x}_i) \quad (1)$$

By repeating  $N$  such layers, the classifier gains expressive power akin to layered neural networks. Each layer can be simplified into a single unitary gate that includes both data encoding and trainable elements.

$$L(i) \equiv U(\theta_i + \mathbf{w}_i \mathbf{x}_i) \quad (2)$$

This further highlighting its parallels with classical artificial neural networks by introducing the concept of *weights*. Although these unitaries can, in principle, be decomposed into elementary rotations covering the entire  $SU(2)$  space ( $R_y R_z R_z$  or  $R_x R_z R_x$ ), the resulting circuits can become quite deep and parameter-heavy.

To address practical limitations of this fully general  $SU(2)$  approach, Skolik *et al.* propose a more hardware-friendly adaptation using multiple qubits and a reduced gate set (Skolik et al., 2022). In particular, they insert the data through a  $R_x$  rotation whose angle is  $\arctan(x \cdot w)$ , with  $w$  being a trainable weight, then introduce trainable parameters through additional single-qubit gates  $R_y$  and  $R_z$ . After applying the rotation gates, the qubits are entangled with their nearest neighbours using a circular entanglement pattern.

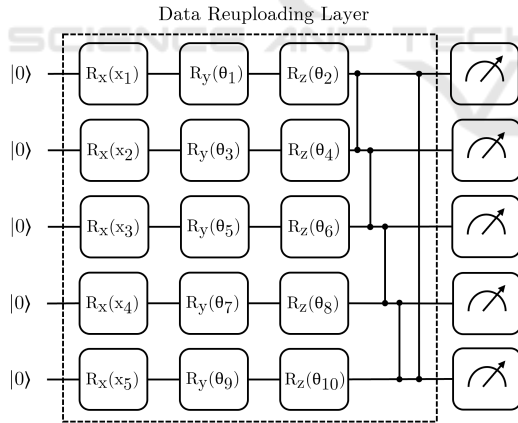


Figure 1: Circuit structure using a single DR layer.

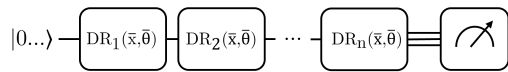


Figure 2: Generic circuit structure using  $N$  DR layers.

Building on this pattern, we define 7 circuits with different numbers of repetitions that will allow us to understand the effects of this technique. Specifically, we define circuits with 1, 2, 3, 4, 5, 10 and 20 repetitions. The circuits have a width of 5 qubits because

this is a sufficient number for our purposes and, given that some of the circuits are relatively deep, a larger width would have consumed a considerable amount of time and computational resources. The resulting circuit is shown in Figure 1 with layered structure shown in Figure 2.

## 2.2.2 Quantum Variational Classifier Model

It is worth noting that we are not training the model, we are just exploring the results of random weights. So the process is as follows:

1. Random data and weights are loaded into the circuit and the vector of states is measured to obtain the exact probabilities of each state.
2. An interpretation function is used to transform state probabilities into class probabilities. Specifically, a modulus function is used, which in binary classification is equivalent to measuring only the bit with the least weight, a location that has been shown to avoid barren plateaus (Cerezo et al., 2021). This is the reason why this function has been chosen over others such as the number of ones in the bit-string or half states for each class.
3. Finally, the cross-entropy function is used to calculate the circuit's loss function as shown in (3).

$$L = -\frac{1}{N} \sum_{j=1}^N \sum_{i=1}^C y_{j,i} \log \hat{y}_{j,i} \quad (3)$$

## 2.2.3 Monte Carlo Configuration

We generate 4,000 random parameter sets following a uniform distribution from  $-\pi$  to  $\pi$ , which is not enough to explore all the possibilities of the circuits. This particularly affects our expressibility metric. For example, for an increment of, say,  $0.1 \pi$ , we would need just over  $10^{13}$  parameters. It is clear that a full scan is not reasonable. We ran the same 4000 random parameter sets for each data and number of repetitions, for a total of 5.6 million circuits.

To calculate the metrics in Section 2.3, instead of storing all the values, we calculate the means and standard deviations of the 4000 circuits for each data. As these statistical measures are subject to some error, we calculate confidence intervals. (4) is the uncertainty of the mean, (5) is the uncertainty of the standard deviation and (6) is the propagation of this uncertainty in the means of these measurements.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}} \quad (4)$$

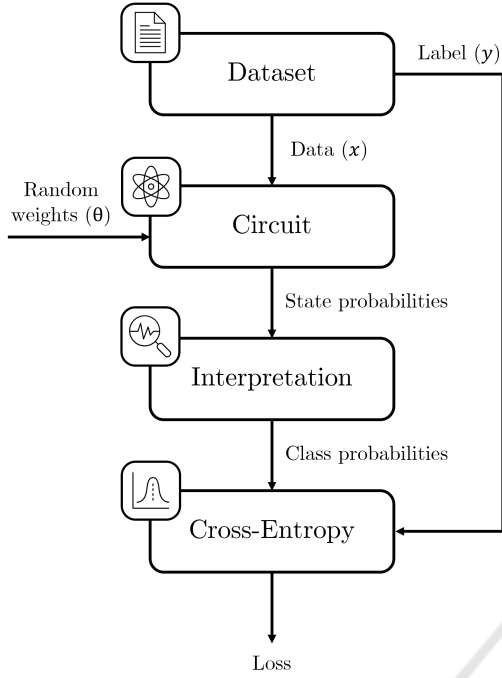


Figure 3: Data Reuploading evaluation process.

$$\sigma_{\sigma} = \frac{\sigma}{\sqrt{2(N-1)}} \quad (5)$$

$$\Delta\bar{x} = \frac{1}{M} \sqrt{\sum_{j=1}^M (\Delta x_j)^2} \quad (6)$$

We also evaluated the minima and maxima; however, due to the intrinsic complexity in quantifying their uncertainties and the limited interpretability of these estimates, we chose not to compute them.

### 2.2.4 Implementation

We adopt Skolik’s variant of data reuploading, using Qiskit’s `TwoLocal` class (Javadi-Abhari et al., 2024), so that it has the same functionality as the standard circuits used for QML, such as the `EfficientSU2`.

We execute the VQCs in a simulator, using the `AerSimulator` class from `qiskit_aer`. We then obtain the exact probabilities using its state vector. A noisy simulation was discarded because it consumed too many resources to get the exact probabilities and these were more desirable than the quasiprobabilities to better appreciate the effect of the technique.

## 2.3 Evaluation Metrics

The performance of a VQC is fundamentally determined by its ability to approximate an optimal solution within a given problem space. This capability

depends not only on the expressiveness of the circuit but also on its optimization properties, which influence how effectively the model can be trained. Specifically, the literature identifies three key characteristics that govern the suitability of a variational model for practical applications:

- *Expressibility* – The extent to which the circuit can explore the entire search space, ensuring that the chosen parametrised quantum states are sufficiently diverse to represent a wide range of solutions.
- *Trainability* – The ability of the circuit to be efficiently optimized, which is closely tied to the presence or absence of barren plateau—regions in the parameter space where gradients vanish, hindering learning.
- *Completeness* – The circuit’s capacity to reach the global minimum of the optimization landscape, indicating its ability to converge toward the most optimal solution rather than getting trapped in local minima.

These properties serve as fundamental evaluation criteria for assessing the effectiveness of VQCs in machine learning tasks. In the following subsections, we provide a detailed discussion of each characteristic, along with the specific metrics used in this work to quantify them.

### 2.3.1 Expressibility

The expressibility of a VQC is defined as the proportion of the Hilbert space it can cover through its parametrised gates (Holmes et al., 2022; Nakaji and Yamamoto, 2021; Sim et al., 2019). In other words, a highly expressible VQC can, in principle, generate a wide range of quantum states, allowing for a more expansive search over the solution space. This characteristic typically depends on factors such as the depth of the circuit, the variety of gate operations, and the data encoding strategy, key elements in Data Reuploading.

However, while high expressibility expands the search space, it also tends to correlate with an increased likelihood of encountering barren plateaus (Larocca et al., 2024). The deeper and more complex a circuit is the more parameters it has to optimize, often leading to highly non-trivial energy landscapes where gradients may vanish. Our methodology therefore quantifies expressibility by computing the Kullback–Leibler (KL) divergence between the distribution of fidelities generated by the VQC and that of the Haar-random ensemble, following the approach in (Sim et al., 2019).



Specifically, we estimate this fidelity distribution by sampling pairs of parameter vectors, computing their corresponding state fidelities, and treating these fidelities as random variables. We state fidelities with (7) (Jozsa, 1994), which can be simplified as the square Bhattacharyya coefficient. Comparing the sampled distribution to that of Haar-random states then yields a KL divergence score. The closer this score is to zero, the more expressible the VQC is. By varying the number of Data Reuploading layers, we can characterize how they affect the circuit’s overall expressibility and, in turn, its subsequent trainability.

$$F(\rho_1, \rho_2) = \left\{ \text{Tr} \left[ (\sqrt{\rho_1} \rho_2 \sqrt{\rho_1})^{\frac{1}{2}} \right] \right\}^2 \quad (7)$$

### 2.3.2 Loss Dispersion as a Trainability Indicator

Trainability in the context of gradient-based optimization refers to the ability of a VQC to avoid barren plateaus—regions in the parameter space where the gradient of the cost function vanishes exponentially with respect to the number of qubits or the circuit depth (McClellan et al., 2018; Cerezo et al., 2021; Larocca et al., 2024). When a circuit experiences barren plateaus, learning becomes infeasible because the parameter updates are effectively driven by infinitesimal gradients. As a result, the training converges extremely slowly or fails to converge altogether.

We focus on barren plateaus because they are the most present problem in the existing literature on the trainability of VQCs. Moreover, it has been shown to occur together with other optimization challenges such as the tightness of the minima of the cost function or the exponential concentration of costs over the mean (Arrasmith et al., 2022). The latter is particularly relevant, not only because it allows us to focus on a single problem to address them all, but also because it allows us to identify barren plateaus through costs instead of gradients, thus making it possible to analyse the trainability of circuit without optimizing it.

To evaluate trainability in our experiments, we initialize the variational parameters with random values and analyse the statistical behaviour of the cost function. Specifically, we track loss dispersion, defined as the standard deviation of cost values across multiple circuit instances. If this standard deviation decreases significantly as the system size increases, it may indicate the presence of a barren plateau. A circuit is considered trainable if it maintains a sufficiently large standard deviation, ensuring that parameter updates remain effective in practice. Conversely, a near-zero standard deviation suggests an exceedingly flat opti-

mization landscape, signalling the presence of a barren plateau.

### 2.3.3 Completeness Estimation

A given VQC is considered complete if it is capable of generating the optimal solution state within the manifold of states it can represent (Holmes et al., 2022). In the context of quantum optimization, this optimal state corresponds to the ground state of a problem Hamiltonian or, more generally, the state that minimizes a given cost function.

While higher expressibility increases the probability that the optimal state is included in the circuit’s accessible state space, it does not guarantee its presence for every problem instance. Completeness therefore captures a fundamental aspect of solution feasibility: even if a circuit is highly expressible, it may still lack the ability to encode the optimal solution. Additionally, achieving optimality requires more than just completeness; trainability is essential to ensure that the optimization process successfully converges to the minimum. Without sufficient trainability, even a complete and expressive VQC may fail to reach its optimal state in practice.

In this work, rather than training the circuit directly or computing the exact reachable minima, we adopt an empirical approach to estimate completeness. We uniformly sample a large number of parameter configurations and evaluate the loss function for each data point in the dataset under these parameters. For each data point, we record the lowest observed loss, interpreting this as the minimum value the circuit can feasibly attain. By averaging these minimum values across the dataset, we estimate the circuit’s apparent global minimum. This method provides a practical indicator of whether the circuit is capable, in principle, of representing solutions with sufficiently low loss, thereby offering an empirical assessment of its completeness.

### 2.3.4 Uniform Classifier

To provide a meaningful reference for evaluating the performance of our Data Reuploading circuits, we introduce what we term the Uniform Classifier. This classifier serves as a baseline by simulating a circuit that applies a Hadamard gate to each qubit, effectively preparing the qubits in an equal superposition of all possible states. Sampling from this circuit results in a uniform distribution over the possible outcomes, representing a scenario where no learning or optimization has taken place. By comparing the performance of our Data Reuploading circuits against this uniform baseline, we can better assess the impact of this strat-

egy on the model’s performance. The Uniform Classifier provides a clear benchmark for understanding the advantages and limitations of Data Reuploading in practical quantum machine learning tasks.

### 3 RESULTS

In this section, we present the experimental results of our systematic evaluation on the impact of Data Reuploading on the metrics. Through a combination of quantitative analysis and visual representations, we aim to offer a comprehensive understanding of the trade-offs associated with this embedding strategy in practical quantum machine learning tasks.

To summarize our findings, we provide Table 1 that presents the average values and corresponding standard errors for all evaluated metrics across different numbers of reuploading layers. Additionally, we include three plots (Figures 4, 5, and 6) that visually depict the trends in expressibility, loss dispersion, and the cross-entropy loss, respectively, as the number of reuploading layers increases. These results collectively illustrate the effects of the embedding technique on the performance and behaviour of VQCs.

#### 3.1 Discussion

In this subsection, we discuss the implications of our experimental results, focusing on the three characteristics that permeate this document: expressibility, trainability and completeness.

##### 3.1.1 Expressibility

The expressibility results, as depicted in Figure 4, indicate that increasing the number of reuploading layers degrades the circuit’s ability to explore a broader portion of the Hilbert space. This is evidenced by the increase in the expressibility metric from 0.4928 for a single layer to 1.1297 for three layers, suggesting a decreased capacity to generate diverse quantum states. This suggests that additional reuploading layers do not necessarily translate into greater state-space coverage.

However, beyond this point, the trend changes. From three layers onward, expressibility values appear to saturate, reaching 1.18 approximately for both 10 and 20 layers. This suggests that further layering may introduce redundancies in parametrisation, effectively constraining the range of achievable states. This phenomenon aligns with prior observations in the literature, where excessive circuit depth has been associated with the concentration of generated states

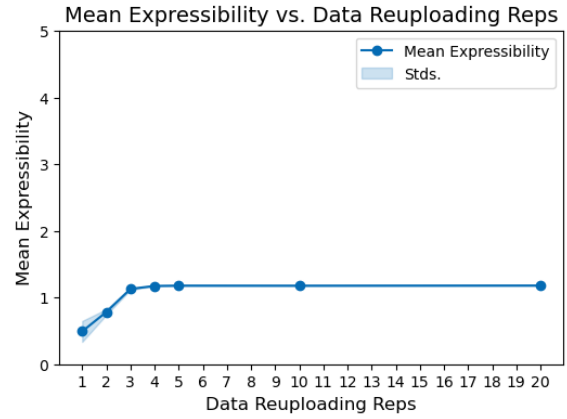


Figure 4: The evolution of the effect of data reuploading layers on the average expressibility. In this case, only a worsening of this metric is observed. With a small increase after two repetitions, a larger increase after three repetitions and a stagnation after the fourth repetition.

in specific subregions of the Hilbert space, thereby limiting the overall diversity of the quantum model.

It is also worth noting that, while expressibility is crucial for ensuring a sufficiently rich state space, excessive layering may lead to diminishing returns and potentially adverse effects. This is reflected in our results, where the expressibility values plateau despite increasing the number of reuploading layers. Such behaviour suggests that, beyond a certain depth, the circuit does not meaningfully expand its expressibility but instead exhibits structural limitations that constrain further improvements.

In comparison to the uniform classifier ( $\ln(150) \approx 5.0106$ ), all configurations employing Data Reuploading achieve significantly lower expressibility values (more expressive), confirming that this approach enables more efficient state-space exploration than a purely random distribution of quantum states. Nevertheless, the observed saturation highlights the need for careful selection of the number of reuploading layers to optimize expressibility without introducing unnecessary complexity.

##### 3.1.2 Trainability

The trainability analysis, based on the loss dispersion metric, reveals a clear trend: as the number of reuploading layers increases, the dispersion of loss values systematically decreases (Figure 5). This reduction suggests that parameter updates become less effective as the circuit depth grows, ultimately leading to a highly constrained optimization landscape.

For a single reuploading layer, the loss dispersion is relatively high at 0.5950, indicating a sufficiently diverse range of loss values across different parameter initializations. This suggests that the optimization

Table 1: Comparison of the effect of Reuploading Layers on Expressivity, Loss dispersion, and Loss metrics. Each of the presented measurements is an average on the 200 rows of data parameters. Alongside the metrics is the standard error, signalling the uncertainty associated to the statistical measures.

| DR<br>Reps            | Avg. Expressibility                  | Avg. Loss Dispersion          | Avg. Mean Loss                | Avg Min. & Max. Loss    |
|-----------------------|--------------------------------------|-------------------------------|-------------------------------|-------------------------|
|                       | Metrics Avg. ( $\pm$ Standard Error) |                               |                               | (Min., Max.)            |
| 1                     | 0.4928 ( $\pm$ 0.0111)               | 0.5949 ( $\pm$ 0.0005)        | 0.5738 ( $\pm$ 0.0007)        | (0.0231, 2.2449)        |
| 2                     | <b>0.7828</b> ( $\pm$ 0.0030)        | <b>0.1884</b> ( $\pm$ 0.0002) | <b>0.3792</b> ( $\pm$ 0.0002) | <b>(0.0558, 1.1299)</b> |
| 3                     | 1.1297 ( $\pm$ 0.0009)               | 0.1079 ( $\pm$ 0.0001)        | 0.3581 ( $\pm$ 0.0001)        | (0.1062, 0.8389)        |
| 4                     | 1.1747 ( $\pm$ 0.0003)               | 0.0901 ( $\pm$ 0.0001)        | 0.3548 ( $\pm$ 0.0001)        | (0.1167, 0.7763)        |
| 5                     | 1.1791 ( $\pm$ 0.0002)               | 0.0898 ( $\pm$ 0.0001)        | 0.3544 ( $\pm$ 0.0001)        | (0.1205, 0.7811)        |
| 10                    | 1.1783 ( $\pm$ 0.0002)               | 0.0905 ( $\pm$ 0.0001)        | 0.3544 ( $\pm$ 0.0001)        | (0.1183, 0.7889)        |
| 20                    | 1.1803 ( $\pm$ 0.0003)               | 0.0905 ( $\pm$ 0.0001)        | 0.3546 ( $\pm$ 0.0001)        | (0.1182, 0.7934)        |
| Uniform<br>Classifier | $\ln(150) \approx 5.0106$            | 0.0                           | $\ln(2)/2 \approx 0.3466$     |                         |

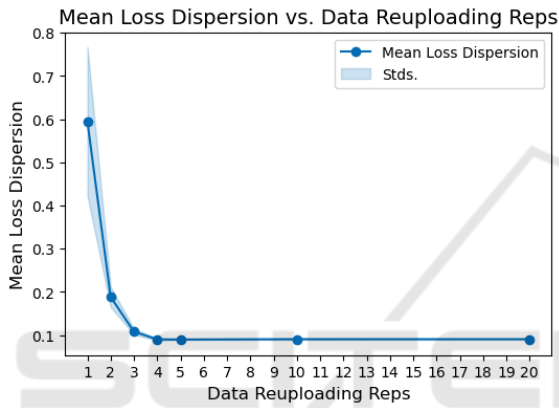


Figure 5: The evolution of the effect of data reuploading layers on the average loss dispersion. This metric drops off abruptly as layers are added. After 4 layers, the values are so low that the circuit is practically untrainable.

landscape retains enough variability to allow effective parameter updates. However, with the introduction of additional layers, loss dispersion steadily declines, dropping to 0.1884 for two layers and further decreasing to 0.0898 by the fifth layer. Beyond this point, the metric plateaus, with values around 0.0905 for circuits with 10 or more layers.

This trend aligns with the onset of barren plateaus, a well-documented issue in deep variational quantum circuits. When the loss dispersion approaches zero, it implies that most parameter configurations yield nearly identical loss values, making it increasingly difficult for gradient-based optimization to distinguish between better and worse configurations. In such cases, the gradient magnitude tends to vanish exponentially with depth, rendering parameter updates ineffective and significantly hindering the training process.

The stagnation of loss dispersion beyond five reuploading layers suggests that, past a certain depth, the circuit reaches a regime where barren plateaus

dominate, making further optimization impractical. This is consistent with prior findings in the literature, which indicate that while increased circuit depth may sometimes enhance expressibility, it simultaneously exacerbates trainability challenges due to gradient decay. In our case, however, no significant improvement in expressibility is observed beyond two reuploading layers. On the contrary, as discussed in Section 3.1.1, further increases in depth appear to hinder even the circuit’s ability to generate diverse quantum states. This suggests that the trade-off between expressibility and trainability is particularly unfavourable for deeper reuploading circuits.

For comparison, the uniform classifier yields a perfectly uniform distribution over possible outcomes, resulting in a loss dispersion of exactly 0.0. This is not due to optimization difficulties but rather a fundamental property of its construction: since it consists solely of Hadamard gates applied to each qubit, it always produces the same probability distribution, regardless of input data. While all reuploading-based circuits maintain non-zero dispersion, the observed decline with increasing layers highlights a fundamental limitation: beyond a certain point, additional reuploading layers do not enhance the circuit’s representation capacity and, instead, severely compromise its trainability.

### 3.1.3 Completeness

The cross-entropy loss, depicted in Figure 6, provides insights into the completeness of the VQCs. As expected, the average loss initially decreases as the number of reuploading layers increases, indicating an improvement in classification performance. This trend is particularly evident when transitioning from one to two layers, where the average loss drops from 0.5738 to 0.3792, suggesting that the circuit gains the ability to represent more accurate decision boundaries.

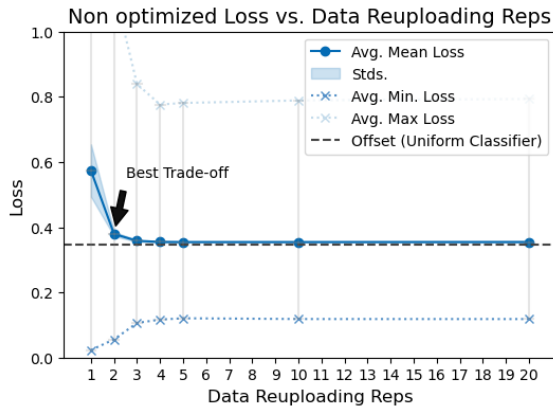


Figure 6: The evolution of the effect of data reuploading layers on the loss. In the average of the mean loss, there is an improvement that decreases until the fifth layer, where there is stagnation. However, the average of the minimums only shows an improvement with two layers, after which it worsens and also ends up stagnating on the fifth layer. Therefore, for our problem, we found the best performance of this technique with two repetitions.

However, beyond two layers, the reduction in loss slows down, and by four layers, it begins to plateau. The difference between five, ten and twenty layers is negligible (0.3544, 0.3544 and 0.3546). This stagnation suggests that additional reuploading layers do not significantly enhance the circuit’s ability to represent lower-loss configurations, reinforcing the findings from the expressibility and trainability analyses.

A crucial observation supporting this interpretation is the behavior of the minimum and maximum loss values across different configurations. Unlike the average loss, which consistently decreases up to two layers, the minimum loss does not follow the same trend. While the lowest observed loss for one layer is 0.0231, this value actually increases slightly for two layers (0.0558) before continuing to rise at deeper depths, reaching 0.1062 for three layers and 0.1205 for five layers. Beyond this point, the minimum loss stabilizes around 0.118 for ten and twenty layers. Similarly, the maximum loss, which is a measure of the worst-case performance, decreases significantly between one and two layers (2.2449 to 1.1299), but beyond this point, the decrease is much more moderate, converging to a stable range (0.7763 to 0.7934).

These results suggest that while increasing the number of reuploading layers initially provides a more expressive circuit capable of better classification, excessive depth introduces diminishing returns. The increase in minimum loss beyond two layers indicates that, despite higher expressibility, the circuit is less likely to encode optimal solutions. This is likely a direct consequence of the barren plateaus observed in Section 3.1.2, where deeper circuits suffer from van-

ishing gradients, concentrating most sampled loss values around the offset—the average loss of the uniform classifier ( $\approx 0.3466$ ). As a result, fewer sampled configurations reach low-loss regions, making optimization increasingly difficult.

Finally, while our Monte Carlo sampling approach provides a practical means of estimating completeness, it is important to recognize its limitations. The minimum and maximum values are drawn from a finite sample of 4,000 random parameter configurations (see Section 2.2.3). Consequently, the absence of lower minimum or higher maximum loss values may partially be a result of sampling limitations. However, the overall trend—where deeper circuits exhibit a narrower range of loss values concentrated around the offset—provides strong supporting evidence for the onset of barren plateaus, making it increasingly difficult for the circuit to find and exploit low-loss configurations.

### 3.1.4 Overall Implications

A key takeaway from our analysis is that the best trade-off in this setting is achieved with two reuploading layers. This configuration achieves the lowest average loss while avoiding the excessive expressibility that could introduce high variance in the optimization landscape. The relatively low maximum loss at two layers (1.1299) further supports this, as it suggests a more controlled and stable learning process compared to deeper circuits. Although the minimum loss at two layers is slightly higher than for a single layer, the fact that it remains relatively low while the average loss is significantly reduced indicates that this configuration is more likely to generalize well across different parameter initializations. This suggests that the circuit remains trainable while also benefiting from an enhanced capacity to generate meaningful decision boundaries.

Additionally, expressibility experiences only a slight increase (less expressive) when moving from one to two layers, indicating that the circuit at this depth remains sufficiently expressive while also being more structured and less prone to redundancy. Beyond two layers, expressibility stagnates, and trainability deteriorates, making further increases in depth counterproductive.

## 4 CONCLUSION

Collectively, these results demonstrate that Data Reuploading is a powerful but nuanced embedding strategy. It is not guaranteed that increasing the number



of reuploading layers will make circuits more expressive, but even if it does not, it can bias the circuit for lower losses without losing too much trainability to the point of being unable to optimise, thus improving classification performance. However, excessive layering introduces significant trainability problems, ultimately leading to barren plateaus.

Moreover, these results are in line with recent literature showing that hardware efficient ansatzes using local costs functions are trainable with controlled depths and suffer from barren plateaus if they are too deep (Cerezo et al., 2021). The circuit used is based on a hardware efficient approach and the interpretation function is local (least significant qubit) for two classes. Thus, we note that the position of the barren plateaus can vary depending on the problem, and factors such as the cost function can delay its appearance.

These findings underscore the importance of carefully balancing the number of reuploading layers to achieve robust and trainable quantum machine learning models. While Data Reuploading enhances the representational power of Variational Quantum Circuits, our results highlight that more layers do not necessarily translate into better performance. Instead, an optimal number must be chosen to avoid barren plateaus and preserve effective optimization dynamics.

#### 4.1 Future Work

This study highlights the potential and effect of Data Reuploading in VQCs for quantum machine learning, yet several avenues remain open for investigation. First, it would be instructive to compare DR based VQCs under different cost functions, datasets, and learning tasks (including multi-class classification and regression) to identify how data characteristics and performance metrics affect our findings. Next, it would be valuable to extend the study to circuits with more qubits, to see if the observed trends change with the width of the circuit.

Given that real quantum devices are prone to noise, assessing the resilience of DR in noisy environments remains vital to determining its practical viability. Finally, exploring alternative DR configurations, such as variations in gate arrangements or parameter-sharing strategies, could uncover novel approaches to optimising performance and scalability.

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## APPENDIX

### Code Availability

The code used to implement the algorithms and perform the analyses in this study is available at the DRCC GitHub repository<sup>1</sup>. The repository includes all relevant scripts, data processing methods, and example datasets necessary to replicate the results presented in this paper. For questions regarding the code or its application, please contact the corresponding author at [danel.a.a@deusto.es](mailto:danel.a.a@deusto.es).

<sup>1</sup>Repository available in <https://github.com/DeustoTech/research-quantum-conference-irqsoft-DRCC>